# 1.4 Working with the Supremum

The least upper bound axiom (Axiom 1.3.1) tells us that suprema exist. To use a supremum, or to recognize one, we need a handle on its second property. As defined (Definition 1.2.4), (S2) quantifies over **all** upper bounds of the set, and in practice there is no way to list them. This section replaces (S2) by an equivalent statement quantified over small positive numbers $\varepsilon$, which can be checked by exhibiting elements. With it we prove the rules for computing suprema of scaled sets and of sums of sets, and meet the first $\varepsilon/2$ argument of the book.

## 1.4.1 The epsilon characterization

Look back at how (S2) was verified for $(0, 1)$ in Example (The supremum need not belong to the set). We supposed that some $M < 1$ was an upper bound and then produced an element of the set above $M$. Writing $M = 1 - \varepsilon$, the argument really showed: for every $\varepsilon > 0$, some element of the set exceeds $1 - \varepsilon$. That is a statement about elements of the set, not about upper bounds, and it is exactly equivalent to (S2).

*A supremum is an upper bound that the set comes arbitrarily close to.*

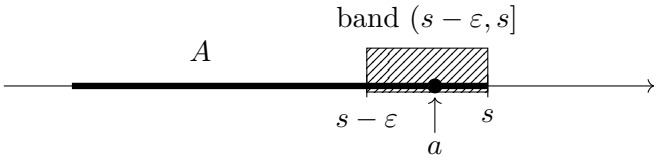
**Theorem 1.4.1. Epsilon Characterization of the Supremum**

Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$ be an upper bound of $A$. Then $s = \sup A$ if and only if **for every** $\varepsilon > 0$ **there exists** $a \in A$ with $a > s - \varepsilon$.

In symbols, for an upper bound $s$ of $A$:

$$s = \sup A \quad \Longleftrightarrow \quad \forall \varepsilon > 0 \;\; \exists a \in A : \; s - \varepsilon < a \; (\leq s).$$

The part in parentheses is automatic, since $s$ is an upper bound; it is written to show where $a$ lies.




*Idea.* Draw the band $(s - \varepsilon, s]$ just below $s$. The theorem says: $s$ is the supremum exactly when every such band, however thin, contains a point of $A$.

($\Rightarrow$) The number $s - \varepsilon$ is smaller than $s$, so it cannot be an upper bound, by (S2). Not being an upper bound means precisely that some element of $A$ lies above it.

($\Leftarrow$) We need (S2): every upper bound $M$ satisfies $s \leq M$. Suppose some upper bound $M$ lies below $s$. Then the band between $M$ and $s$ contains no point of $A$, and the hypothesis says every band does. So the band of width $s - M$, that is, the hypothesis with $\varepsilon = s - M$, gives the contradiction.

*Proof.* ($\Rightarrow$) Suppose $s = \sup A$, and let $\varepsilon > 0$. Then $s - \varepsilon < s$ by Theorem 1.1.2 (a). If $s - \varepsilon$ were an upper bound of $A$, then (S2) would give $s \leq s - \varepsilon$, contradicting $s - \varepsilon < s$ by trichotomy. So $s - \varepsilon$ is not an upper bound of $A$, which means there is $a \in A$ with $a > s - \varepsilon$.

($\Leftarrow$) Suppose that for every $\varepsilon > 0$ there is $a \in A$ with $a > s - \varepsilon$. Since $s$ is an upper bound of $A$ by hypothesis, (S1) holds, and it remains to check (S2). Let $M$ be an upper bound of $A$, and suppose, for a contradiction, that $M < s$. Put $\varepsilon := s - M$, which is positive by Theorem 1.1.2 (a). By hypothesis there is $a \in A$ with $a > s - \varepsilon = M$. This contradicts $M$ being an upper bound of $A$. Hence $s \leq M$ by trichotomy, which is (S2). So $s = \sup A$.

■

The theorem does not use the axiom. It holds in any ordered field, and it **recognizes** a supremum; the axiom is what guarantees that there is one to recognize. The two halves are used differently. The forward direction is how a known supremum is **used**: it hands us elements of $A$ as close to $\sup A$ as we like. The backward direction is how a candidate is **confirmed**: it replaces "every upper bound is at least $s$" by "every band below $s$ meets $A$", which can be checked by writing down an element.

*Remark.* The hypothesis "$s$ is an upper bound" cannot be dropped. For $A = (0, 1)$ and $s = \frac{1}{2}$, every band $(\frac{1}{2} - \varepsilon, \frac{1}{2}]$ meets $A$, yet $\frac{1}{2}$ is not the supremum. The band condition alone says only that $s$ is not too big; being an upper bound says it is not too small.

The element $a$ that the theorem produces **depends on** $\varepsilon$, as the order "$\forall\varepsilon\ \exists a$" allows. For $A = (0, 1)$ and $s = 1$, the choice $a := \max\{\frac{1}{2}, 1 - \frac{\varepsilon}{2}\}$ works: it lies in $(0, 1)$ and exceeds $1 - \varepsilon$. For $\varepsilon = \frac{1}{10}$ it gives $a = \frac{19}{20}$; for $\varepsilon = \frac{1}{1000}$ it must be above $\frac{999}{1000}$, and $\frac{19}{20}$ no longer works. No single element of $(0, 1)$ works for every $\varepsilon$.

*Warning.* **The witness depends on $\varepsilon$.** The theorem gives, **for each** $\varepsilon$, **some** $a$; it never gives one $a$ that serves every $\varepsilon$ at once. A proof that fixes $a$ first and then says "for every $\varepsilon > 0$, $a > s - \varepsilon$" has swapped the quantifiers, and for $(0, 1)$ it asserts something false. When you take a witness from the theorem, write "choose $a \in A$ (depending on $\varepsilon$) with ...".

*Quick check.* Let $s$ be an upper bound of a set $A \subseteq \mathbb{R}$. Compare

$$\text{(i)}\ \ \forall\varepsilon > 0\ \ \exists a \in A : a > s - \varepsilon \qquad \text{and} \qquad \text{(ii)}\ \ \exists a \in A\ \ \forall\varepsilon > 0 : a > s - \varepsilon.$$

Statement (i) says $s = \sup A$. What does (ii) say, and when is it true?

*Solution.* Fix $a \in A$. The condition "$a > s - \varepsilon$ for every $\varepsilon > 0$" forces $a \geq s$: if $a < s$, then $\varepsilon := s - a > 0$ would give $a > s - (s - a) = a$, which is absurd. Since $s$ is an upper bound, also $a \leq s$, so $a = s$. Conversely, if $s \in A$, then $a = s$ satisfies $s > s - \varepsilon$ for every $\varepsilon > 0$.

So (ii) says exactly that $s \in A$, that is, $s$ is the **maximum** of $A$. Swapping the quantifiers turned "supremum" into "maximum". For $A = (0, 1)$ and $s = 1$, (i) is true and (ii) is false.

Here is the theorem at work on a set where the witness must be found by a short computation. The Idea finds it backwards; the solution writes it forwards.

**Example. Confirming a supremum with epsilon**

Prove that $\sup\{x^2 : x \in (0,1)\} = 1$.

*Solution. Idea.* The upper bound part is easy. For the band condition, fix $\varepsilon > 0$; we need $x \in (0,1)$ with $x^2 > 1-\varepsilon$. Try $x = 1-h$ with $h > 0$ small. Then $x^2 = 1-2h+h^2 > 1-2h$, so it suffices that $1-2h \geq 1-\varepsilon$, that is, $h \leq \frac{\varepsilon}{2}$. But we also need $x \in (0,1)$, that is, $0 < h < 1$, and $\frac{\varepsilon}{2}$ can be large. Capping it fixes this: take $h = \min\{\frac{1}{2}, \frac{\varepsilon}{2}\}$.

*Proof.* Let $A = \{x^2 : x \in (0,1)\}$. For $x \in (0,1)$, Theorem 1.1.2 (g) gives $x^2 < 1^2 = 1$, so 1 is an upper bound of $A$.

Let $\varepsilon > 0$, and let $h \coloneqq \min\{\frac{1}{2}, \frac{\varepsilon}{2}\}$, which depends only on $\varepsilon$. Then $0 < h \leq \frac{1}{2}$, so $x \coloneqq 1-h$ lies in $[\frac{1}{2}, 1) \subseteq (0,1)$, and $x^2 \in A$. Since $h > 0$, $h^2 > 0$, and so

$$x^2 = 1 - 2h + h^2 > 1 - 2h \geq 1 - \varepsilon,$$

where the last step uses $h \leq \frac{\varepsilon}{2}$. By Theorem 1.4.1, $\sup A = 1$.

The $\frac{1}{2}$ in the minimum is arbitrary: any cap below 1 keeps $x$ positive. What matters is that a constraint discovered late in the Idea (here $h < 1$) is built into the constant from the start of the proof.

## 1.4.2 Sup approximation

The forward direction of Theorem 1.4.1 is used so often with $\varepsilon = \frac{1}{n}$ that it deserves its own statement.

**Corollary 1.4.2. Sup Approximation**

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above. For every $n \in \mathbb{Z}_{>0}$ there is $a_n \in A$ with

$$\sup A - \frac{1}{n} < a_n \leq \sup A.$$

*Proof.* By Axiom 1.3.1, $\sup A$ exists. Let $n \in \mathbb{Z}_{>0}$. Then $n > 0$, and $\frac{1}{n} > 0$ by Theorem 1.1.2 (f). By Theorem 1.4.1 ($\Rightarrow$) with $\varepsilon = \frac{1}{n}$, there is $a_n \in A$ with $a_n > \sup A - \frac{1}{n}$, and $a_n \leq \sup A$ by (S1).

■

The corollary gives, for each $n$ separately, an element $a_n$. Choosing one for every $n$ at once, so as to form a sequence $(a_n)$, is a use of the axiom of countable choice; we flag it and move on. Once limits are defined in Chapter 2, this says that $\sup A$ is the limit of a sequence from $A$. We call the move **sup approximation**: "$\sup A - \frac{1}{n}$ is not an upper bound, so some $a_n \in A$ exceeds it."

## 1.4.3 Monotonicity and scaling

We now compute suprema of sets built from other sets. The first rule says that enlarging a set cannot lower its supremum.

**Theorem 1.4.3. Monotonicity of the Supremum**

Let $A \subseteq B \subseteq \mathbb{R}$, where $A$ is nonempty and $B$ is bounded above. Then $\sup A$ and $\sup B$ exist, and $\sup A \leq \sup B$.

*Proof.* Since $A$ is nonempty and $A \subseteq B$, the set $B$ is nonempty; as it is bounded above, $\sup B$ exists by Axiom 1.3.1. Every $a \in A$ lies in $B$, so $a \leq \sup B$ by (S1) for $B$. Hence $\sup B$ is an upper bound of $A$. So $A$ is nonempty and bounded above, $\sup A$ exists by Axiom 1.3.1, and $\sup A \leq \sup B$ by (S2) for $A$. ■

Strict inclusion does not give strict inequality: $(0, 1) \subsetneq (0, 1]$, and both have supremum 1. The hypothesis that $A$ is nonempty cannot be dropped, since $\emptyset \subseteq B$ always and $\emptyset$ has no supremum.

For scaling we need the scaled set. For $A \subseteq \mathbb{R}$ and $c \in \mathbb{R}$, let

$$cA := \{ca : a \in A\}.$$

For example $2(0, 1] = (0, 2]$ and $(-1)A = -A$.

**Theorem 1.4.4. Scaling the Supremum**

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$.

(a) If $c > 0$, then $cA$ is nonempty and bounded above, and $\sup(cA) = c \sup A$.

(b) If $c < 0$, then $cA$ is nonempty and bounded below, and $\inf(cA) = c \sup A$.

*Idea.* For (a), prove the equality as two inequalities. Multiplying the inequality $a \leq \sup A$ by $c > 0$ shows that $c \sup A$ is an upper bound of $cA$, which gives $\leq$. For $\geq$, undo the scaling: dividing by $c$ turns an upper bound of $cA$ into an upper bound of $A$. No $\varepsilon$ is needed, because multiplication by $c > 0$ is invertible and preserves order. For (b), a negative factor reverses order, so write $c = -|c|$ and let Theorem 1.3.2 handle the sign.

*Proof.* Let $s := \sup A$, which exists by Axiom 1.3.1. Since $A$ is nonempty, so is $cA$.

(a) Let $c > 0$. For every $a \in A$, $a \leq s$ by (S1), so $ca \leq cs$ by Theorem 1.1.2 (c) (or trivially if $a = s$). Hence $cs$ is an upper bound of $cA$, so $\sup(cA)$ exists by Axiom 1.3.1, and $t := \sup(cA) \leq cs$ by (S2). For the reverse inequality, let $a \in A$. Then $ca \in cA$, so $ca \leq t$. Multiplying by $\frac{1}{c} > 0$ (Theorem 1.1.2 (f) and (c)) gives $a \leq \frac{t}{c}$. Hence $\frac{t}{c}$ is an upper bound of $A$, and (S2) gives $s \leq \frac{t}{c}$. Multiplying by $c > 0$ gives $cs \leq t$. By trichotomy, $\sup(cA) = cs$.

(b) Let $c < 0$, so $-c > 0$. The set $-(cA) = \{-ca : a \in A\}$ equals $(-c)A$, which by (a) is nonempty and bounded above with $\sup((-c)A) = -cs$. By Theorem 1.3.2 (a),

$cA = -((-c)A)$ is bounded below, and by Theorem 1.3.2,

$$\inf(cA) = -\sup(-(cA)) = -\sup((-c)A) = -(-cs) = cs.$$

This proves the theorem.

■

For $c = 0$ the set $cA$ is $\{0\}$, and $\sup(0A) = 0 = 0 \cdot \sup A$ as well. Part (a) genuinely needs $c > 0$: with $c = -1$ and $A = (0, 1]$, the set $cA = [-1, 0)$ has supremum 0, not $c \sup A = -1$. Part (b) says what happens instead: a negative factor exchanges supremum and infimum, and indeed $\inf[-1, 0) = -1$.

## 1.4.4 The supremum of a sum

For $A, B \subseteq \mathbb{R}$, let

$$A + B := \{a + b : a \in A,\ b \in B\}.$$

For instance $(0, 1) + \{5\} = (5, 6)$ and $[0, 1] + [0, 2] = [0, 3]$. It is natural to expect $\sup(A + B) = \sup A + \sup B$. This time the reverse inequality cannot be obtained by undoing an operation: a single element of $A + B$ mixes one element of each set. We need the $\varepsilon$ characterization.

**Theorem 1.4.5. Supremum of a Sum**

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then $A + B$ is nonempty and bounded above, and

$$\sup(A + B) = \sup A + \sup B.$$

*Try it first (ten minutes): the inequality $\leq$ takes one line. Try to prove $\geq$.*

*Idea.* The attempt that comes first to mind takes $a = \sup A$ and $b = \sup B$: then $a+b \in A+B$, and (S1) for $\sup(A + B)$ gives $\geq$ at once. It fails because a supremum need not belong to its set: for $A = B = (0, 1)$, the number 2 is not in $A + B = (0, 2)$. (The last exercise of this section takes that attempt apart.) The repair is to use elements that are only **close** to the suprema, which is what Theorem 1.4.1 ($\Rightarrow$) supplies.

So we prove the equality as **two inequalities**.

($\leq$) Every $a + b$ is at most $\sup A + \sup B$, so the right side is an upper bound, and the supremum is at most any upper bound.

($\geq$) Work backwards. To show that $\sup(A + B)$ is at least $\sup A + \sup B$, it suffices to find, for each $\varepsilon > 0$, an element of $A + B$ above $\sup A + \sup B - \varepsilon$. The natural candidate is $a + b$ with $a$ close to $\sup A$ and $b$ close to $\sup B$. If $a$ falls short of $\sup A$ by less than $\varepsilon/2$, and $b$ falls short of $\sup B$ by less than $\varepsilon/2$, the total shortfall is less than $\varepsilon$. So we split the budget $\varepsilon$ into two halves, one for each set, and ask Theorem 1.4.1 for an element within $\varepsilon/2$ of each supremum. This is the $\varepsilon/2$ **argument**. The number $\varepsilon/2$ depends only on $\varepsilon$; the elements $a$ and $b$ depend on $\varepsilon$. *Before reading on: the proof asks for elements within $\varepsilon/2$ of each supremum. What would it give if it asked for $\varepsilon$ instead, and would that still be enough?*

*Proof.* By Axiom 1.3.1, $\sup A$ and $\sup B$ exist. Choosing $a_0 \in A$ and $b_0 \in B$ gives $a_0 + b_0 \in A + B$, so $A + B$ is nonempty.

($\leq$) Let $a \in A$ and $b \in B$. By (S1), $a \leq \sup A$ and $b \leq \sup B$, so $a + b \leq \sup A + \sup B$ by Theorem 1.1.2 (b). Hence $\sup A + \sup B$ is an upper bound of $A + B$. By Axiom 1.3.1, $\sup(A + B)$ exists, and by (S2), $\sup(A + B) \leq \sup A + \sup B$.

($\geq$) Let $\varepsilon > 0$. Then $\frac{\varepsilon}{2} > 0$. By Theorem 1.4.1 ($\Rightarrow$), there are $a \in A$ and $b \in B$ (both depending on $\varepsilon$) with

$$a > \sup A - \tfrac{\varepsilon}{2} \quad \text{and} \quad b > \sup B - \tfrac{\varepsilon}{2}.$$

Adding, by Theorem 1.1.2 (b), gives $a + b > \sup A + \sup B - \varepsilon$. Since $a + b \in A + B$, (S1) gives

$$\sup(A + B) \geq a + b > \sup A + \sup B - \varepsilon.$$

This holds for every $\varepsilon > 0$. If we had $\sup(A + B) < \sup A + \sup B$, then the choice $\varepsilon := \sup A + \sup B - \sup(A + B) > 0$ would give $\sup(A + B) > \sup(A + B)$, which is absurd. Hence $\sup(A + B) \geq \sup A + \sup B$.

The two inequalities and trichotomy give $\sup(A + B) = \sup A + \sup B$.

■

The answer to the prompt: with $\varepsilon$ in place of $\varepsilon/2$ the proof gives $\sup(A+B) > \sup A+\sup B-2\varepsilon$ for every $\varepsilon > 0$, and that is still enough, since $2\varepsilon$ runs through all positive numbers as $\varepsilon$ does. Halving is a convenience: it makes the final inequality come out with exactly $\varepsilon$, which is the form Theorem 1.4.1 ($\Leftarrow$) asks for.

The halves of the proof use the two halves of the definition. The first inequality used only that each supremum is an upper bound (S1) and that $\sup(A + B)$ is least (S2). The second used that each of $\sup A$ and $\sup B$ is least, in its $\varepsilon$ form, and that $\sup(A+B)$ is an upper bound. We call the pattern **equality by two inequalities**: prove $\leq$ by exhibiting an upper bound, and $\geq$ with an $\varepsilon$.

Every hypothesis is needed. If $B = \emptyset$, then $A + B = \emptyset$ has no supremum. If $B$ is not bounded above, neither is $A + B$ (Exercise (C2: Exit check: sums of sets)). And the analogous rule for differences is **not** $\sup(A - B) = \sup A - \sup B$: the correct version subtracts an infimum (Exercise (B3: Supremum of a difference)).

*Warning.* **Proving one inequality only.** Here is a tempting "proof" of Theorem 1.4.5: "For $a \in A$ and $b \in B$, $a + b \leq \sup A + \sup B$. Therefore $\sup(A + B) = \sup A + \sup B$." The first sentence proves only that $\sup A + \sup B$ is **an** upper bound of $A + B$, which gives $\leq$. It says nothing about being the **least** one: the same sentence with $\sup A + \sup B + 1$ is equally true, and would "prove" $\sup(A + B) = \sup A + \sup B + 1$. Whenever a supremum is claimed to equal something, both (S1) and (S2), or both inequalities, must appear.

## 1.4.5 Exercises

### 1.4.5.1 A. Check your understanding

**Exercise. A1**

(a) Use Theorem 1.4.1 to confirm that $\sup[0,5) = 5$. Give the witness $a$ as a formula in $\varepsilon$.

(b) Use Theorem 1.4.1 to confirm that $\sup\{3\} = 3$. Does your witness depend on $\varepsilon$? Why not?

(c) Use Theorem 1.4.1 to confirm that $\sup((-1,1) \cup (2,4)) = 4$.

(d) Let $s$ be an upper bound of $A$. Write in symbols, using the $\varepsilon$ characterization, the statement "$s$ is **not** the supremum of $A$".

(e) Run the $(\geq)$ half of the proof of Theorem 1.4.5 on $A = (0,1)$, $B = \{5\}$, and $\varepsilon = \frac{1}{10}$: give an $a$ and a $b$ the proof could use, and check the inequality it derives.

(f) In the proof of Theorem 1.4.5, which line can no longer be justified if $B$ is allowed to be empty?

*Solution.*

(a) 5 is an upper bound of $[0,5)$. Given $\varepsilon > 0$, let $a := \max\{0, 5 - \frac{\varepsilon}{2}\}$. Then $a \geq 0$, and $a < 5$ because both 0 and $5 - \frac{\varepsilon}{2}$ are less than 5; so $a \in [0,5)$. Also $a \geq 5 - \frac{\varepsilon}{2} > 5 - \varepsilon$. By Theorem 1.4.1, $\sup[0,5) = 5$.

(b) 3 is an upper bound of $\{3\}$, and for every $\varepsilon > 0$ the element $a = 3$ satisfies $3 > 3 - \varepsilon$. The witness does not depend on $\varepsilon$ because 3 belongs to the set: by the Quick check above, a single witness for all $\varepsilon$ exists exactly when the supremum is a maximum.

(c) Every element is less than 4, so 4 is an upper bound. Given $\varepsilon > 0$, let $a := \max\{3, 4 - \frac{\varepsilon}{2}\}$. Then $3 \leq a < 4$, so $a \in (2,4)$, and $a \geq 4 - \frac{\varepsilon}{2} > 4 - \varepsilon$. By Theorem 1.4.1 the supremum is 4.

(d) $\exists \varepsilon > 0 \;\; \forall a \in A : a \leq s - \varepsilon$. In words: some band below $s$ contains no point of $A$, so $s - \varepsilon$ is a smaller upper bound.

(e) Here $\sup A = 1$, $\sup B = 5$, and $\frac{\varepsilon}{2} = \frac{1}{20}$. The proof needs $a \in (0,1)$ with $a > 1 - \frac{1}{20} = \frac{19}{20}$, for instance $a = \frac{24}{25}$, and $b \in B$ with $b > 5 - \frac{1}{20}$, which forces $b = 5$. Then $a + b = \frac{149}{25} = 5.96 > 5.9 = \sup A + \sup B - \varepsilon$. Any $a \in (\frac{19}{20}, 1)$ works; the one $b$ works for every $\varepsilon$ because 5 is the maximum of $B$.

(f) The first line, "By Axiom 1.3.1, $\sup A$ and $\sup B$ exist". The axiom needs $B$ nonempty, and $\emptyset$ has no supremum. The line that chooses $b \in B$ in the $(\geq)$ half fails too, since there is nothing to choose.

### 1.4.5.2 B. Practice

**Exercise. B1: A computed supremum**

Prove that $\sup\{x^2 : x \in (-1,2)\} = 4$. Hence show that the set has no maximum.

*Solution.* Let $A = \{x^2 : x \in (-1,2)\}$. *Upper bound.* Let $x \in (-1,2)$. If $x \geq 0$, then $x < 2$ gives $x^2 < 4$ by Theorem 1.1.2 (g). If $-1 < x < 0$, then $|x| < 1$ by Proposition 1.1.3 (d), so $x^2 = |x|^2 < 1 < 4$ by Theorem 1.1.2 (g). So 4 is an upper bound, and $4 \notin A$.

*Band condition.* Working backwards as in Example (Confirming a supremum with epsilon): with $x = 2 - h$ we have $x^2 = 4 - 4h + h^2 > 4 - 4h$, so $h \leq \frac{\varepsilon}{4}$ suffices, provided $0 < h < 3$ to

keep $x \in (-1,2)$. So let $\varepsilon > 0$ and $h := \min\{1, \frac{\varepsilon}{4}\}$, which depends only on $\varepsilon$. Then $0 < h \le 1$, so $x := 2 - h \in [1,2) \subseteq (-1,2)$, and since $h^2 > 0$,

$$x^2 = 4 - 4h + h^2 > 4 - 4h \ge 4 - \varepsilon.$$

By Theorem 1.4.1, $\sup A = 4$. Since $4 \notin A$, the set has no maximum, by Exercise (B1: Maxima are suprema) (a).

**Exercise. B2: Supremum of a union**

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Prove that $A \cup B$ is bounded above and $\sup(A \cup B) = \max\{\sup A, \sup B\}$.

*Solution.* Let $m := \max\{\sup A, \sup B\}$; both suprema exist by Axiom 1.3.1. If $x \in A \cup B$, then $x \le \sup A \le m$ or $x \le \sup B \le m$, so $m$ is an upper bound of $A \cup B$, which is nonempty. By symmetry we may assume $m = \sup A$. Given $\varepsilon > 0$, Theorem 1.4.1 ($\Rightarrow$) gives $a \in A$ with $a > \sup A - \varepsilon = m - \varepsilon$, and $a \in A \cup B$. By Theorem 1.4.1 ($\Leftarrow$), $\sup(A \cup B) = m$.

**Exercise. B3: Supremum of a difference**

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and $B \subseteq \mathbb{R}$ nonempty and bounded below. Prove that

$$\sup\{a - b : a \in A,\ b \in B\} = \sup A - \inf B.$$

Give an example showing that the right side cannot be replaced by $\sup A - \sup B$.

*Solution.* The set in question is $A + (-B)$. By Theorem 1.3.2 (a), $-B$ is bounded above, and it is nonempty. By Theorem 1.3.3, $\inf B = -\sup(-B)$, that is, $\sup(-B) = -\inf B$. So Theorem 1.4.5 applies to $A$ and $-B$:

$$\sup(A + (-B)) = \sup A + \sup(-B) = \sup A - \inf B.$$

For the example, take $A = \{0\}$ and $B = [0,1]$. The differences $0 - b$ fill $[-1,0]$, whose supremum is $0 = \sup A - \inf B$, whereas $\sup A - \sup B = -1$.

### 1.4.5.3 C. Going deeper

**Exercise. C1: Products of sets**

For $A, B \subseteq \mathbb{R}$ let $AB := \{ab : a \in A,\ b \in B\}$.

(a) Show that $\sup(AB) = \sup A \cdot \sup B$ can fail for nonempty bounded sets $A, B$.

(b) Prove that it holds when $A$ and $B$ are nonempty, bounded above, and contained in $(0, \infty)$.

*Solution* (Hint 1). For (a), try sets of negative numbers. For (b), the inequality $\leq$ is routine; for $\geq$, fix one element $b \in B$ and compare the set $bA$ with $AB$.

*Solution* (Hint 2). For $\geq$ in (b): fix $b \in B$. Which theorem computes $\sup(bA)$, and which compares the suprema of two sets, one inside the other? What does the inequality you get say about $b$, once $b$ varies?

*Solution.*

(a) Take $A = B = [-2, -1]$. Then $\sup A \cdot \sup B = (-1)(-1) = 1$, but $4 = (-2)(-2) \in AB$, so $\sup(AB) \geq 4 \neq 1$ by (S1).

(b) Let $s := \sup A$ and $t := \sup B$, which exist by Axiom 1.3.1. Since $A$ contains a positive element and $s$ is an upper bound, $s > 0$; likewise $t > 0$.

($\leq$) For $a \in A$ and $b \in B$, we have $0 < a \leq s$ and $0 < b \leq t$, so $ab \leq sb \leq st$ by Theorem 1.1.2 (c) (with trivial equality cases). So $st$ is an upper bound of the nonempty set $AB$, $\sup(AB)$ exists, and $\sup(AB) \leq st$ by (S2).

($\geq$) Fix $b \in B$. Then $bA \subseteq AB$ and $b > 0$, so Theorem 1.4.4 and Theorem 1.4.3 give $bs = \sup(bA) \leq \sup(AB)$. Multiplying by $\frac{1}{s} > 0$ gives $b \leq \frac{\sup(AB)}{s}$. As $b \in B$ was arbitrary, $\frac{\sup(AB)}{s}$ is an upper bound of $B$, so $t \leq \frac{\sup(AB)}{s}$ by (S2), that is, $st \leq \sup(AB)$.

By trichotomy, $\sup(AB) = st$. Positivity was used in ($\leq$), to multiply inequalities, and in ($\geq$), to scale by $b$ and divide by $s$; part (a) shows it cannot be dropped.

**Exercise. C2: Exit check: sums of sets**

Let $A, B \subseteq \mathbb{R}$ be nonempty.

(a) Assume $A$ and $B$ are bounded above. Without looking back, prove $\sup(A + B) = \sup A + \sup B$. This time, instead of proving $\geq$ separately, show that $\sup A + \sup B$ satisfies both hypotheses of Theorem 1.4.1 for the set $A + B$.

(b) Assume $A$ and $B$ are bounded below. State and prove the corresponding formula for $\inf(A + B)$.

(c) Show that if $B$ is not bounded above, then neither is $A + B$.

*Solution.*

(a) Let $s := \sup A + \sup B$, where both suprema exist by Axiom 1.3.1. *Upper bound:* for $a \in A$, $b \in B$, (S1) and Theorem 1.1.2 (b) give $a + b \leq s$. *Band condition:* let $\varepsilon > 0$; then $\frac{\varepsilon}{2} > 0$, and Theorem 1.4.1 ($\Rightarrow$) gives $a \in A$ and $b \in B$, depending on $\varepsilon$, with $a > \sup A - \frac{\varepsilon}{2}$ and $b > \sup B - \frac{\varepsilon}{2}$. Adding, $a + b > s - \varepsilon$, and $a + b \in A + B$. By Theorem 1.4.1 ($\Leftarrow$), $s = \sup(A + B)$.

(b) $\inf(A + B) = \inf A + \inf B$. Indeed $-(A + B) = (-A) + (-B)$, since $-(a + b) = (-a) + (-b)$. By Theorem 1.3.2 (a), $-A$ and $-B$ are nonempty and bounded above,

so (a) and Theorem 1.3.3 give

$$\begin{aligned}\inf(A+B) &= -\sup((-A)+(-B)) \\ &= -\sup(-A) - \sup(-B) = \inf A + \inf B.\end{aligned}$$

Here $A+B$ is bounded below because $-(A+B)$ is bounded above, by (a) and Theorem 1.3.2 (a).

(c) Fix $a_0 \in A$ and let $M \in \mathbb{R}$. Since $B$ is not bounded above, $M - a_0$ is not an upper bound of $B$, so some $b \in B$ has $b > M - a_0$. Then $a_0 + b \in A + B$ and $a_0 + b > M$. As $M$ was arbitrary, $A + B$ is not bounded above.

**Marking scheme.** (10 points) - 1: (a) says why $\sup A$, $\sup B$ exist and why $A + B$ is nonempty. - 2: (a) shows $\sup A + \sup B$ is an upper bound of $A + B$, adding the two (S1) inequalities. - 3: (a) fixes an arbitrary $\varepsilon > 0$, obtains $a \in A$ and $b \in B$, **depending on** $\varepsilon$, within $\varepsilon/2$ of their suprema, and concludes with Theorem 1.4.1 ($\Leftarrow$). - 2: (b) reduces to (a) through $-(A+B) = (-A)+(-B)$ and Theorem 1.3.3, including why $A+B$ is bounded below. - 2: (c) produces, for an arbitrary $M$, an element of $A+B$ above $M$. Fatal errors (0 for the step): taking $a = \sup A \in A$; choosing one $a$ before $\varepsilon$ is fixed; proving only $\leq$ in (a). A solution of (a) that proves $\leq$ and $\geq$ separately, as in Theorem 1.4.5, earns the 5 points of the second and third lines if the $\geq$ half is complete. A solution of (b) that repeats the argument of (a) with lower bounds earns the (b) points.

**Exercise. C3: Find the wrong line**

Each "proof" below contains a line that is false in general. Find it, give a set for which it fails, and decide whether the claim itself is true.

(a) *Claim.* If $A \subseteq \mathbb{R}$ is nonempty and bounded above and $a < 1$ for every $a \in A$, then $\sup A < 1$. *"Proof."* (1) Let $s = \sup A$. (2) Since $s \in A$, the hypothesis gives $s < 1$. (3) Hence $\sup A < 1$.

(b) *Claim.* If $A, B \subseteq \mathbb{R}$ are nonempty and bounded above, then $\sup(A+B) \geq \sup A + \sup B$. *"Proof."* (1) Let $a = \sup A$ and $b = \sup B$. (2) Then $a \in A$ and $b \in B$, so $a+b \in A+B$. (3) By (S1), $a + b \leq \sup(A + B)$, which is the claim.

*Solution.*

(a) Line (2) is wrong: it assumes $\sup A \in A$, which fails for $A = (0,1)$, where $\sup A = 1 \notin A$. For that set the claim is **false**: every element is less than 1, and $\sup A = 1 \not< 1$. The correct conclusion is $\sup A \leq 1$, which follows from (S2) because 1 is an upper bound.

(b) Line (2) is wrong for the same reason: for $A = B = (0,1)$, neither supremum belongs to its set, so $a+b = 2$ need not lie in $A+B = (0,2)$, and here it does not. The claim is **true**, since it is half of Theorem 1.4.5, but this argument does not prove it; a correct argument for this inequality is the $\varepsilon/2$ argument in the proof of Theorem 1.4.5. The flawed line is valid exactly when $A$ and $B$ have maxima, which is the situation the author of the "proof" silently imagined.